

Patterns in Play — Module 04: Cognition — Think About It

Part A: Family Discussion Questions

1. What is "Strategy"? (A plan for how to win or solve a problem in a game).
2. Why do you get better at games the more you play them? (Your brain builds a better model of the game).
3. What is the difference between luck and skill in a game?
4. How does sorting toys by color, size, or type help your brain? (Categories make the world easier to navigate).
5. Can you think three moves ahead in a game like checkers? That is called "temporal depth."
6. Why is it important to change your strategy when the old one stops working? (Cognitive flexibility!).
7. What is a "Pattern"? Can you spot patterns in your favorite game?
8. Why are puzzles good for your brain? (They require matching, sorting, and problem-solving).
9. How do you decide between two choices in a game? What goes on inside your head?
10. Is it better to think fast or think carefully? (It depends on the situation!).

Part B: Hands-On Activities and Reflection

1. Play a strategy game together (checkers, chess, Connect Four). After each move, say your thinking out loud: "I moved here because..." Making thinking visible helps everyone learn.
2. Sort a collection of objects (buttons, Legos, coins) by one rule, then re-sort by a different rule. Your brain just switched categories — that is cognitive flexibility!
3. Do a jigsaw puzzle together. Notice your strategy: Do you start with edges? Group by color? Everyone's brain uses a different approach.
4. Play "Spot the Pattern": Take turns creating simple patterns with objects (red, blue, red, blue...). Can the other person continue the pattern?

5. Design a maze on paper. Trade mazes with a family member and solve each other's. Did the designer predict where the solver would get stuck?
6. Play "20 Questions": Think of an object. Others can only ask yes/no questions. This game trains your brain to sort, categorize, and narrow down possibilities.
7. Build the tallest tower you can with playing cards. When it falls, discuss: What worked? What failed? Adjust your strategy and try again.
8. Play a game you have never played before. Pay attention to how your brain learns the rules. At first it is confusing, then patterns emerge. That is your model building!
9. Invent a "Thinking Game" for your family: Create a challenge that requires brainpower (a riddle, a code, a logic puzzle). Test it on everyone.
10. After playing today, reflect: What was the best strategy you used? What would you do differently next time? Good thinkers review their thinking.